SQL Quest — Administrator Guide

SQL Quest · Administrator Guide · 2026-11-05

Administrators manage accounts and roles, decide which content is visible, configure badges and game rules, and look after backups. To install or update the platform itself, see the Installation Guide.

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1. Roles

Every account has exactly one role. In the server version the server checks the role on **every request**, so nobody can use features outside their role.

	Student	Teacher	Admin
Lessons, challenges, bosses, XP, badges	✅	preview	preview
Join a class, peer review, team missions	✅		
See own progress	✅	✅	✅
Create classes, assignments, teams		✅ own classes	✅ all
Class analytics and student progress		✅ own classes	✅ all
Create lessons and challenges		✅ own	✅ all
Manage users and roles, reset passwords			✅
Hide/publish any challenge, badges, settings, backups			✅

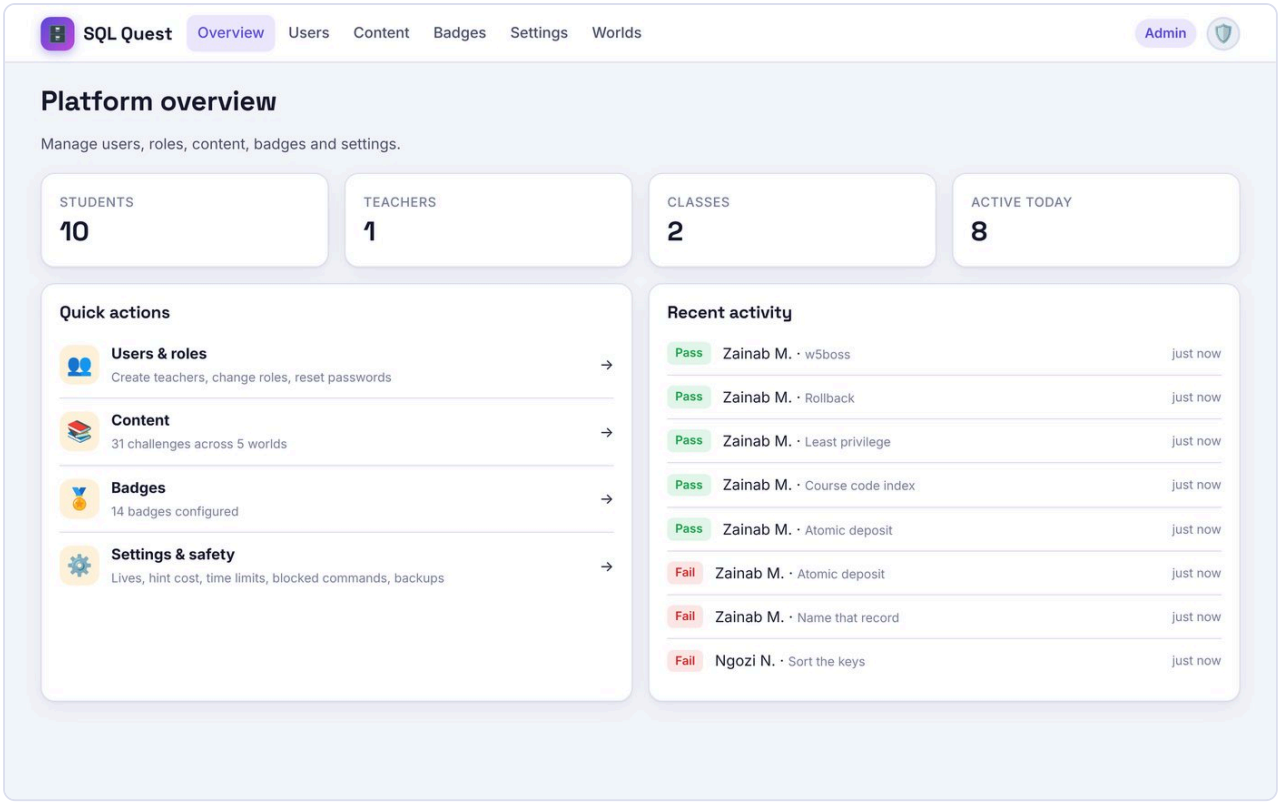
Students never see other students' answers or private details. Leaderboards show only display name, avatar, XP, level, streak and badge count.

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2. First steps on a new installation

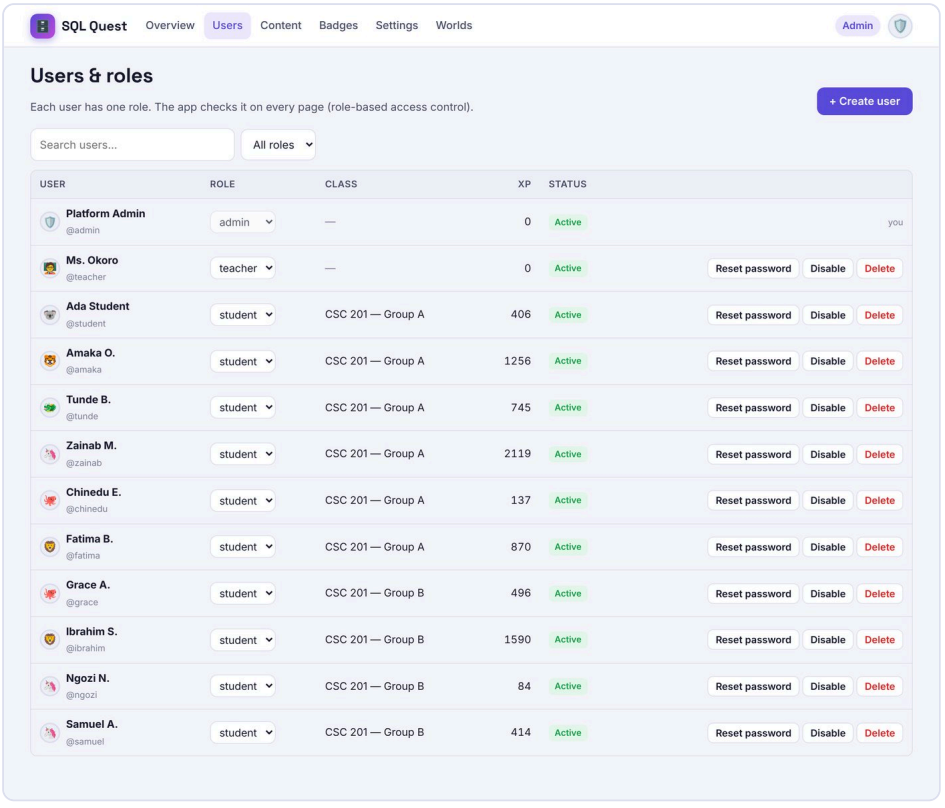
1. Log in with the admin username and password set during installation (`ADMIN_USERNAME` / `ADMIN_PASSWORD`).
2. Account menu → **Profile & settings** → **Change password**.
3. **Settings**: check that **Show demo accounts on the login page** is off, and decide whether to **Allow students to sign up**.
4. **Users** → **+ Create user** for each teacher (role *teacher*, with a temporary password they change at first login).

If the platform was started with demo data (`SEED_DEMO=true`), delete the demo accounts or start again with an empty database before real use. The demo passwords are public.



3. Users and roles

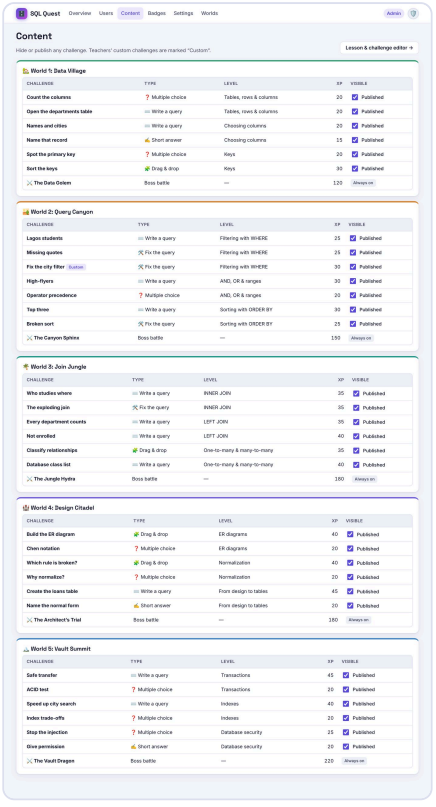
Users lists every account. Use the search box and role filter to find people.



Action	Effect
Role drop-down	Changes the role immediately. You can't change your own role, so there is always at least one admin.
Reset password	Sets a new password and signs the user out everywhere. Tell them the new password; they can change it in their profile.
Disable / Enable	A disabled account can't log in and is signed out at once. Its data is kept.
Delete	Permanently removes the account, its progress, submissions, badges and shared queries. Classes it taught are moved to you. **This can't be undone**; export a backup first if unsure.
+ Create user	Creates an account with any role.

4. Content

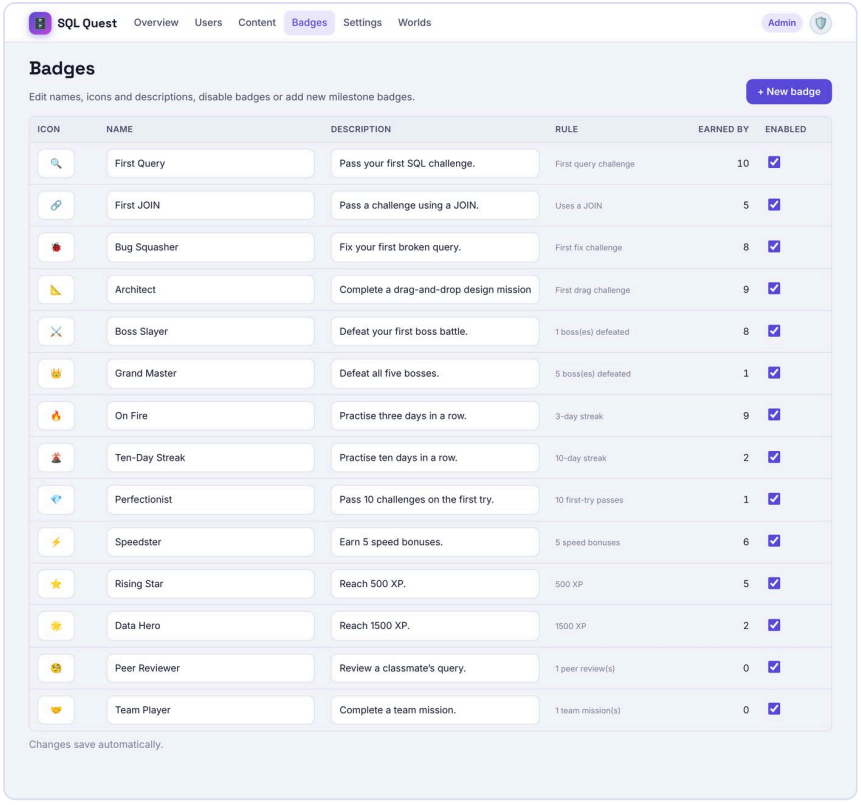
Content lists every challenge in every world, including teachers' custom ones (marked *Custom*). Untick **Visible** to hide a challenge from students; tick it again to publish. Boss battles are always on. To write or edit content, use **Lesson & challenge editor →** (the same editor teachers use).



5. Badges

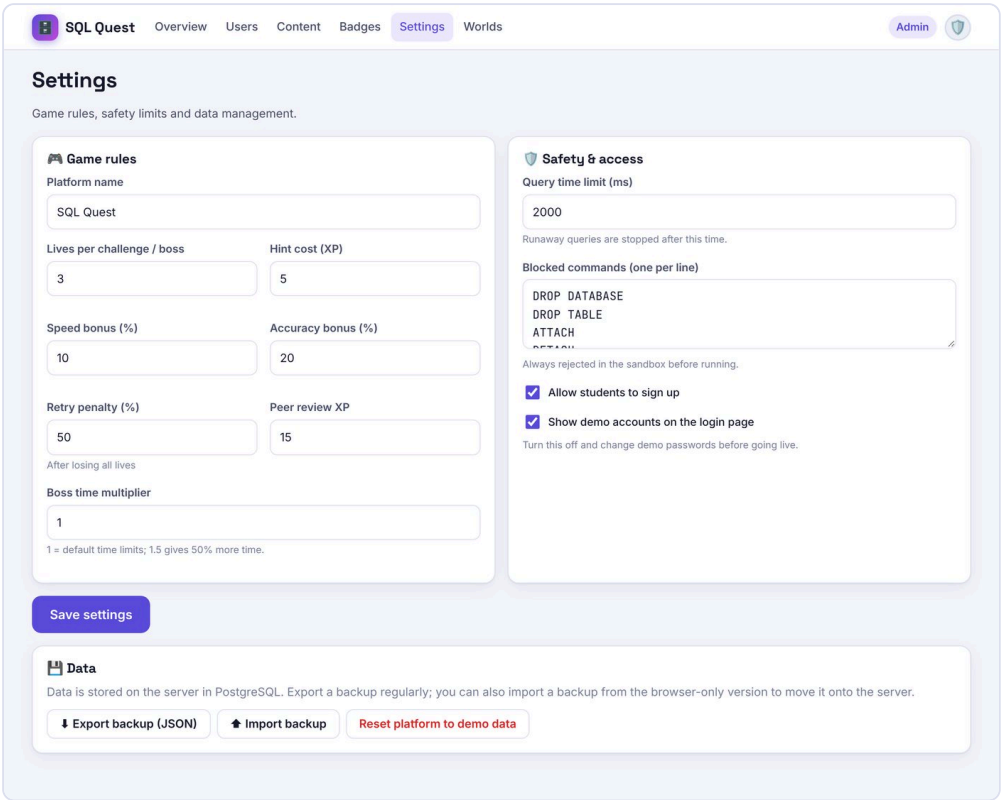
Badges lets you change each badge's **icon** (an emoji), **name** and **description**, and switch it on or off. Changes save as soon as you leave the field.

+ New badge creates a milestone badge awarded automatically when a student reaches a value of: XP, challenges completed, best streak, bosses defeated, first-try passes or peer reviews given. Students who already qualify receive it straight away.



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6. Settings



Game rules

Setting	Default	Meaning
Platform name	SQL Quest	Shown in the top bar and on the login page.
Lives per challenge / boss	3	Wrong answers allowed before "Out of lives".
Hint cost (XP)	5	XP taken for each hint.
Speed bonus (%)	10	Extra XP for passing within the par time.
Accuracy bonus (%)	20	Extra XP for passing with no wrong attempts.
Retry penalty (%)	50	XP removed if the student lost all lives before passing.
Peer review XP	15	XP for each review written.
Boss time multiplier	1	1.5 gives 50% more time in every boss battle; useful for younger or newer classes.

Safety and access

Setting	Default	Meaning
Query time limit (ms)	2000	Queries running longer are stopped.
Blocked commands	DROP DATABASE, DROP TABLE, ATTACH, DETACH, PRAGMA, VACUUM, LOAD_EXTENSION, TRUNCATE	Always refused before running.
Allow students to sign up	on	Off = only admins create accounts.
Show demo accounts on the login page	off on new servers	Turn **off** on any real site.

Every query runs in a brand-new copy of a small practice database, so even allowed statements can't affect real data.

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7. Backups and moving data

At the bottom of **Settings**:

- ⬇ **Export backup (JSON)** downloads everything: accounts (with securely hashed passwords), progress, classes, custom content and settings. Keep it somewhere safe; it contains personal data.
- ⬆ **Import backup replaces all current data** with a backup file. Everyone is signed out afterwards.
- **Moving from the browser-only version to the server:** export a backup in the browser version, log in to the server as admin, and import it. Passwords keep working and are upgraded to stronger hashing at each user's next login.

- **Reset platform to demo data** only appears when the server allows it (`ALLOW_DEMO_RESET=true`). Never enable it on a live site.

For a server, also back up the PostgreSQL database regularly (see the Installation Guide, section 6).

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8. Security checklist

- ☐ Strong, unique admin password; at least two admin accounts in case one is locked out.
- ☐ Demo accounts hidden and deleted (or never seeded).
- ☐ Site served over **HTTPS**, with `COOKIE_SECURE=true` and `TRUST_PROXY=true` behind a proxy.
- ☐ `ALLOW_DEMO_RESET` left off.
- ☐ Regular database backups, stored securely.
- ☐ Disable accounts of people who leave.

Built-in protections: scrypt password hashing, HTTP-only session cookies, rate limits on login and sign-up (repeated failures are blocked for 15 minutes), checks against cross-site requests, answers never sent to browsers, and server-side marking.

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9. Troubleshooting

Problem	Fix
"Too many attempts" at login	Wait 15 minutes. Repeated failed logins are blocked for that time.
A user can't log in	Check they are **Enabled** in Users; reset their password.
"Could not save your change"	The database could not be reached. Check that PostgreSQL is running; the change is kept and retried on the next save.
Students see "The SQL Quest server is not responding"	The server process or database is down. Check the server logs (`docker compose logs app`).
Locked out of every admin account	Someone with server access **stops the app** and runs `node server/scripts/reset-password.js <username> <new-password> --make-admin` with the same `DATABASE_URL`, then starts the app again (with Docker: `docker compose stop app`, then `docker compose run --rm app node server/scripts/reset-password.js admin NewPass123 --make-admin`, then `docker compose start app`). Prevent this by keeping two admin accounts.